

Experiment: DSP Resource Exploration Using Multiply-Accumulate (MAC) Unit in Vivado HLS

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Objective

To implement a Multiply-Accumulate (MAC) operation using Vitis HLS and analyze DSP resource utilization on FPGA.

Problem Statement

Given three integer inputs A, B, and C, perform the following operation:

[Output = (A B) + C]

Implement the design using Vitis HLS and analyze the generated hardware resources.

Pre-Lab Questions

Q1. What is a DSP Slice in an FPGA?

A DSP Slice is a dedicated hardware block inside an FPGA used for high-speed arithmetic operations such as multiplication, addition, subtraction and Multiply-Accumulate (MAC). It performs these operations faster than LUT-based implementations.

Q2. Which arithmetic operation typically utilizes DSP resources?

Multiplication (and MAC operations) primarily use DSP slices.

Q3. Calculate the output for:

A	B	C	Output = (A×B)+C
5	4	2	22
10	3	5	35
7	8	1	57

Q4. Why are multipliers expensive when implemented using only LUTs?

Because multiplication requires many logic gates and routing resources. It increases LUT usage, delay and power consumption.

Q5. What is the advantage of using dedicated DSP blocks?

- Faster computation
 - Less LUT utilization
 - Lower power consumption
 - Higher operating frequency
 - Efficient for signal processing
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Manual Calculation Questions

Q6. Compute the following MAC operations manually:

1. $(12 \times 5) + 8 = (60+8) = 68$
 2. $(15 \times 4) + 10 = (60+10) = 70$
 3. $(25 \times 3) + 12 = (75+12) = 87$
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HLS Analysis Questions

Q7. What are the major stages performed in Vitis HLS?

1. C/C++ Coding
2. C Simulation
3. C Synthesis
4. RTL Generation
5. RTL Co-Simulation
6. IP Export

Q8. What files are generated after synthesis?

- Verilog/VHDL RTL files
 - Synthesis Report
 - Resource Utilization Report
 - Timing Report
 - IP Package files
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Q9. Observe the RTL Viewer and identify:

- Multiplier block
- Adder block
- Registers
- Input/Output ports

Summary Synthesis Report - mac

General Information
Estimated Quality of Results
Timing Estimate
Performance & Resource Estimates
HW Interfaces
Other Ports
TOP LEVEL CONTROL
SW I/O Information
Top Function Arguments
SW-to-HW Mapping
Pragma Report
Bind Op Report
User Config Op
Bind Storage Report
User Config Storage

HW Interfaces

Other Ports

PORT	MODE	DIRECTION	BITWIDTH
A	ap_none	in	32
B	ap_none	in	32
C	ap_none	in	32
ap_return		out	32

TOP LEVEL CONTROL

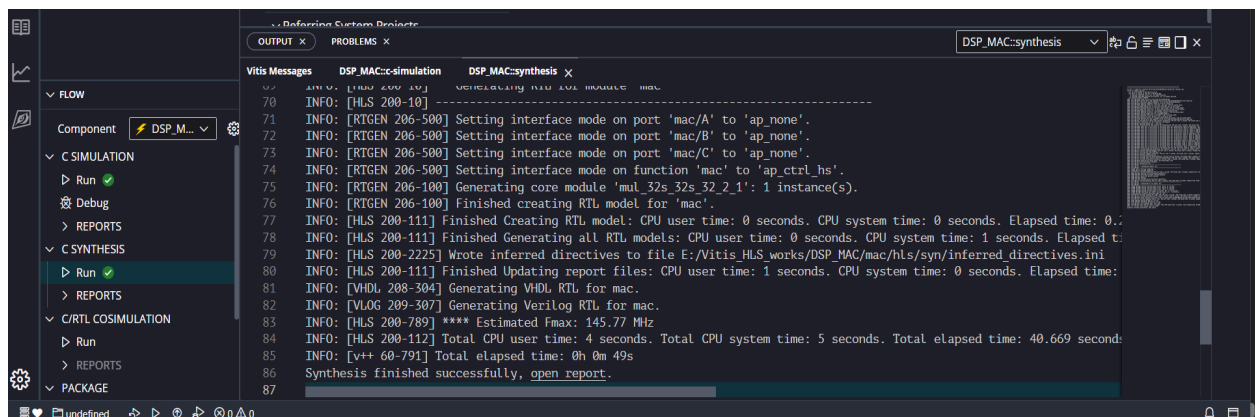
INTERFACE	TYPE	PORTS
ap_clk	clock	ap_clk
ap_rst	reset	ap_rst
ap_ctrl	ap_ctrl_hs	ap_done ap_idle ap_ready ap_start

SW I/O Information
Top Function Arguments

OUTPUT x PROBLEMS x DSP_MAC::synthesis

Resource Analysis

C Synthesis:



After synthesis:

Resource	Value
LUT	106
FF	200
BRAM	0
DSP	3

Q10. Which operation in the design is responsible for DSP utilization?

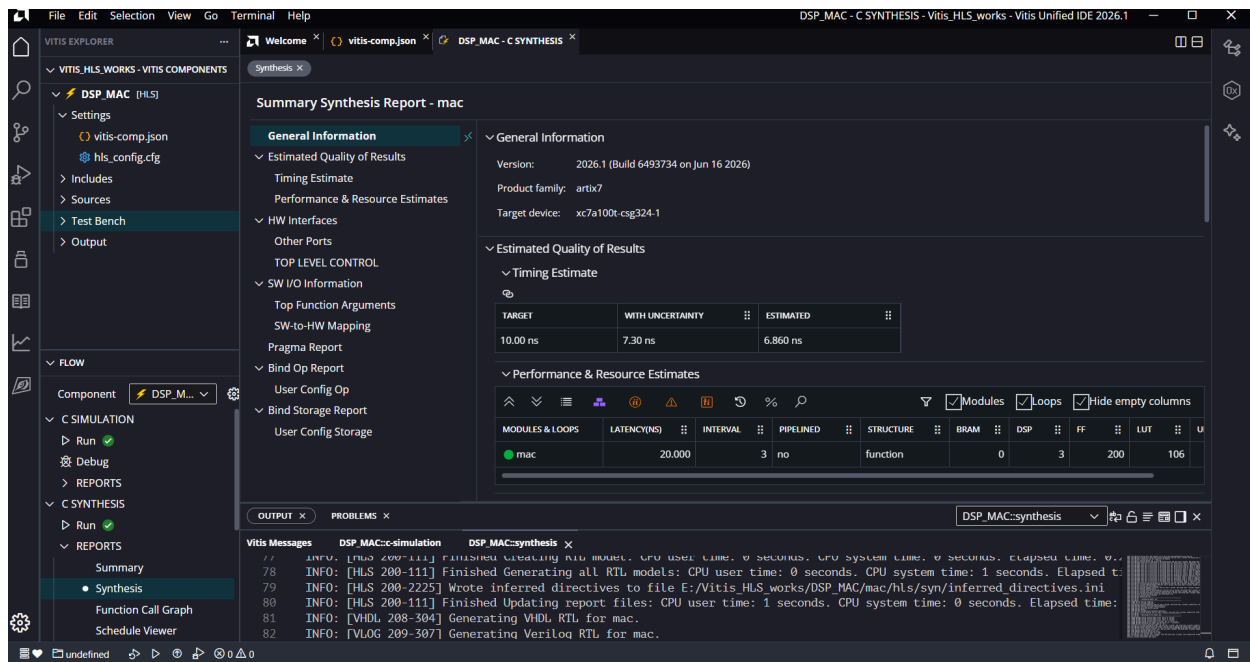
The multiplication operation ($A \times B$) is responsible for DSP utilization. The addition is combined with the multiplier during hardware optimization.

Q11. Why is BRAM utilization expected to be low or zero in this experiment?

No memory arrays or buffers are used in the MAC design. Therefore, Block RAM (BRAM) is not required.

Performance Analysis

Parameter	Value
Target Clock	10 ns
Estimated Clock	6.860 ns
Latency	20 ns
Interval	3



Q12. Compare the resource utilization of:

Design A

[Output = A + B]

Design B

[Output = (A B) + C]

Which design utilizes DSP slices?

Design B utilizes DSP slices because it contains a multiplication operation.

Q13. If the design is modified as:

[Output = A B + C D]

What changes are expected in DSP utilization?

Expected changes:

- DSP utilization will increase because two multiplication operations are present.
- LUT and FF utilization may also increase slightly.
- The overall hardware complexity will be higher.

Output Verification

C Simulation:

Input Set 1

A = 10

B = 5

C = 3

Expected Output = $(8 \times 7) + 4$

= $56 + 4$

= 60

Input Set 2

A = 8

B = 7

C = 4

Expected Output = $(8 \times 7) + 4$

= $56 + 4$

= 60

Input Set 3

A = 12

B = 6

C = 9

Expected Output = $(12 \times 6) + 9$

= 72 + 9

= 81

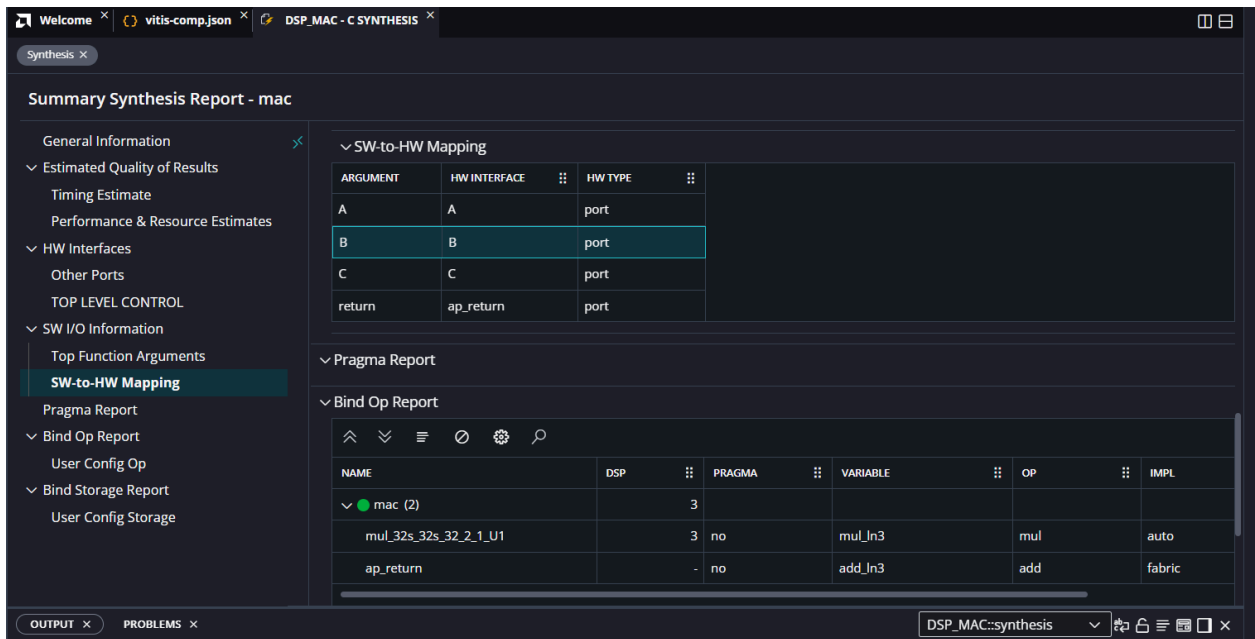
Post-Lab Analysis Questions

Q14. Which FPGA resource increased significantly compared to the Image Negative experiment?

DSP utilization increased because the design contains multiplication, which is mapped onto dedicated DSP slices.

Q15. Explain the relationship between multiplication operations and DSP utilization.

FPGA synthesis tools map multiplication operations to dedicated DSP slices to achieve higher speed and lower LUT usage compared to implementing multipliers entirely in logic fabric.



Summary Synthesis Report - mac

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Timing Estimate

Performance & Resource Estimates

HW Interfaces

Other Ports

TOP LEVEL CONTROL

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Top Function Arguments

SW-to-HW Mapping

Pragma Report

Bind Op Report

ARGUMENT	HW INTERFACE	HW TYPE
A	A	port
B	B	port
C	C	port
return	ap_return	port

NAME	DSP	PRAGMA	VARIABLE	OP	IMPL
mac (2)	3				
mul_32s_32s_2_1_U1	3	no	mul_ln3	mul	auto
ap_return	-	no	add_ln3	add	fabric

Q16. Mention three FPGA applications where DSP slices are heavily used.

Applications using DSP slices:

- FIR Filters
 - FFT Processors
 - CNN Accelerators
 - Audio Signal Processing
 - Image Processing
 - Wireless Communication Systems
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